STACK IT UP PRESENTS

THE SUPPLEMENT FIELD GUIDE

19 science-backed supplements. Dosing, timing, trade-offs — decoded.

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stackitup - your stack, dialed in.

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MULTIVITAMIN

Covers baseline micronutrient gaps from day-to-day eating.

DOSE	TIMING	ONSET
1 capsules	Afternoon	Subtle; most people don't feel a multi day-to-day — benefit is cumulative nutrient adequacy

BENEFITS

- Fills everyday gaps in vitamins A, C, D, E, K, and most B vitamins
- Insurance policy on days your diet is less than ideal
- Supports baseline energy metabolism and immune function
- Convenient single dose vs stacking many individual vitamins

HOW IT WORKS

A multivitamin delivers the RDA or near-RDA dose of micronutrients your body can't make on its own. Water-soluble vitamins (B, C) are excreted if you overshoot; fat-soluble (A, D, E, K) are stored, so stick to reasonable doses.

DOSE NOTES

One serving per day with food. Avoid mega-dose formulas — more is rarely better and can push fat-soluble vitamins into the toxic range.

TIMING NOTES

Take with the largest meal of the day so fat-soluble vitamins (A, D, E, K) absorb properly.

GOOD FOR

- Anyone with an inconsistent diet
- Restrictive diets (elimination, low-calorie)
- Frequent travelers

AVOID IF

- You already take high-dose individual vitamins
- You have hemochromatosis (skip iron-containing formulas)
- You're on warfarin (vitamin K interacts — talk to your doctor)

SIDE EFFECTS

- Nausea if taken on empty stomach
- Bright yellow urine (harmless — excess B2 riboflavin)
- Constipation or upset stomach from iron in some formulas

COMMON FORMS

Capsule / tablet
Most common; cheapest

Gummy
Palatable but usually misses iron, magnesium, potassium

Methylated B
Better absorption for MTHFR variants

OMEGA-3 (EPA/DHA)

Supports heart, brain, and joint health.

DOSE	TIMING	ONSET
1–2 g	Morning	4–12 weeks for measurable changes in blood omega-3 index

BENEFITS

- Lowers triglycerides and supports cardiovascular health
- Anti-inflammatory effects for joints and recovery
- DHA is a structural lipid in brain and eye tissue
- May support mood and reduce symptoms of mild depression

HOW IT WORKS

EPA and DHA are long-chain omega-3 fatty acids that incorporate into cell membranes and compete with pro-inflammatory omega-6 pathways. They're the precursors to resolvins and protectins — signaling molecules that actively resolve inflammation.

DOSE NOTES

Aim for 1–2 g combined EPA+DHA daily. Check the label — a '1000 mg fish oil softgel' often contains only 300 mg actual EPA+DHA.

TIMING NOTES

Take with a meal containing fat for best absorption. Morning is fine; split doses if taking 2+ g to reduce fishy burps.

GOOD FOR

- People who eat fatty fish less than twice a week
- Elevated triglycerides
- Joint stiffness
- Vegans (via algae-based DHA/EPA)

AVOID IF

- You're on blood thinners (talk to your doctor — mild additive effect)
- Scheduled surgery in the next week
- Fish or shellfish allergy (use algae oil instead)

SIDE EFFECTS

- Fishy aftertaste or burps
- Mild GI upset at high doses
- Slightly increased bleeding time at 3+ g/day

COMMON FORMS

Triglyceride (TG) form
Better absorbed than ethyl ester

Ethyl ester (EE)
Cheapest; slightly lower absorption

Algae oil
Vegan source of EPA/DHA

Krill oil
Phospholipid-bound; smaller doses, pricier

STACKS WITH

ASHWAGANDHA (KSM-66)

Adaptogen shown to reduce cortisol and perceived stress.

DOSE	TIMING	ONSET
600 mg	Evening	4–8 weeks for full stress and sleep benefits

BENEFITS

- Reduces cortisol and perceived stress in multiple RCTs
- May improve sleep quality and reduce anxiety symptoms
- Modest boost to strength and recovery in training studies
- Some evidence for testosterone support in men

HOW IT WORKS

Ashwagandha is an adaptogen — it modulates the HPA axis (the body's stress response system), blunting cortisol spikes. Active compounds are withanolides.

DOSE NOTES

600 mg/day of KSM-66 or 250–500 mg of Sensoril (standardized extracts). Raw root powder needs much higher doses to match.

TIMING NOTES

Evening is traditional and supports sleep. Splitting AM/PM also works.

GOOD FOR

- Chronic stress or high-stress job
- Poor sleep tied to racing mind
- Athletes training hard
- Men with low-normal testosterone

AVOID IF

- You have hyperthyroidism (it can raise T3/T4)
- You have autoimmune disease (immune-modulating)
- You're pregnant
- You're on sedatives, thyroid meds, or immunosuppressants

SIDE EFFECTS

- Drowsiness
- Mild GI upset
- Rare: liver issues (stop if jaundice, dark urine, or abdominal pain)

COMMON FORMS

KSM-66
Most-studied extract; full-spectrum root

Sensoril
Higher withanolide %, lower dose

Root powder
Traditional; needs higher doses, taste is strong

STACKS WITH

CAFFEINE + L-THEANINE

Smoother energy and focus without the jitters.

DOSE	TIMING	ONSET
100 mg + 200 mg	Morning	20–45 minutes

BENEFITS

- Smoother, more focused energy than caffeine alone
- Reduces jitters, anxiety, and heart-pounding feeling
- Improves attention and reaction time
- Synergy is well-documented in cognitive performance studies

HOW IT WORKS

Caffeine blocks adenosine receptors (reducing sleepiness) and increases dopamine and norepinephrine. L-Theanine, an amino acid from tea, increases alpha brain waves and modulates glutamate — smoothing caffeine's edge without reducing its benefits.

DOSE NOTES

Classic ratio is 1:2 caffeine to theanine — 100 mg + 200 mg is a great starting point. Some stack up to 200 mg + 400 mg.

TIMING NOTES

Morning or early afternoon. Avoid within 8–10 hours of bedtime — caffeine half-life is ~5 hours.

GOOD FOR

- Deep-focus work
- People who get anxious from coffee
- Pre-workout focus
- Studying or creative work

AVOID IF

- You're sensitive to caffeine or have arrhythmia
- You're pregnant (keep caffeine under 200 mg/day)
- You have anxiety disorders that worsen with stimulants
- It's evening — you'll wreck your sleep

SIDE EFFECTS

- Jitters at higher caffeine doses
- Sleep disruption if taken too late
- Tolerance over time — cycle off periodically
- Caffeine withdrawal headache if stopped cold

COMMON FORMS

Combo capsule
Pre-dosed 100/200 mg — convenient

Separate caffeine + theanine
Flexible ratio tuning

Matcha or green tea
Naturally contains both — gentler dose

STACKS WITH

COLLAGEN PEPTIDES

Supports connective tissue, joints, and skin.

DOSE	TIMING	ONSET
10 g	Morning	4–12 weeks for skin and joint benefits

BENEFITS

- Supports skin elasticity and hydration (modest but real effect)
- May reduce joint pain in active people and older adults
- Can support hair and nail growth
- Easy protein boost (~9 g per 10 g serving)

HOW IT WORKS

Oral collagen is broken down to amino acids and small peptides (hydroxyproline-rich) that signal fibroblasts to upregulate collagen synthesis in skin and joint tissue. It's a targeted signal, not direct replacement.

DOSE NOTES

10 g hydrolyzed collagen peptides daily. Joint-focused studies often use 10–15 g type II collagen.

TIMING NOTES

Anytime. Pair with vitamin C to support collagen synthesis. Morning in coffee is a common routine.

GOOD FOR

- Skin aging concerns
- Joint stiffness from training or age
- People who don't eat bone broth or skin-on meats
- Recovering from tendon/ligament injury

AVOID IF

- Fish/beef/pork allergies (depending on source)
- You're strictly vegan (collagen is always animal-derived)

SIDE EFFECTS

- Mild fullness or GI upset
- Rare: allergic reactions

COMMON FORMS

Hydrolyzed peptides (Type I & III)
Skin, hair, nails — mixes easily

Undenatured Type II (UC-II)
Joints — small daily dose (40 mg)

Bone broth
Whole-food alternative; variable collagen content

STACKS WITH

vitc | whey | creatine

FAQ

Marine vs bovine collagen — which is better?

CREATINE MONOHYDRATE

Most-researched performance supplement — strength, power, and cognition.

DOSE	TIMING	ONSET
5 g	Anytime	2–4 weeks to fully saturate muscles; optional 5-day loading phase of 20 g/day speeds it up

BENEFITS

- Increases strength and power output by 5–15%
- Supports lean muscle mass gains over time
- Improves high-intensity exercise capacity
- Emerging evidence for cognition, especially under sleep deprivation
- Most-studied sports supplement — extensive safety record

HOW IT WORKS

Creatine is stored in muscle as phosphocreatine, which rapidly regenerates ATP during short, intense efforts. Supplementation saturates these stores beyond what you get from diet.

DOSE NOTES

5 g daily, every day. Loading (20 g/day for 5 days) is optional — same end result, just faster. No cycling needed.

TIMING NOTES

Anytime. Post-workout with carbs may slightly boost uptake but consistency matters far more than timing.

GOOD FOR

- Anyone doing resistance training
- High-intensity interval athletes
- Vegetarians/vegans (low baseline muscle creatine)
- Older adults for muscle preservation

AVOID IF

- You have significant kidney disease (talk to your doctor)
- You have creatine sensitivity (rare — GI discomfort)

SIDE EFFECTS

- 2–4 lb weight gain from water inside muscle cells (not fat, not bloat)
- Mild GI upset if you take too much at once
- Does NOT cause hair loss, kidney damage, or dehydration in healthy people

COMMON FORMS

Monohydrate
Gold standard — cheapest, most researched, most effective

Micronized monohydrate
Same thing, finer powder — mixes better

HCl, ethyl ester, buffered
Marketed as better absorbed — evidence does not support paying more

GLUCOSAMINE + CHONDROITIN

Well-studied for joint comfort and cartilage support.

DOSE	TIMING	ONSET
1500 mg	Afternoon	2–3 months — slow responder; judge after 12 weeks

BENEFITS

- May reduce joint pain in mild-to-moderate osteoarthritis
- Supports cartilage integrity over time
- Modest anti-inflammatory effect at joint tissue
- Best evidence is for knee OA

HOW IT WORKS

Glucosamine is a precursor to glycosaminoglycans that make up cartilage. Chondroitin inhibits enzymes that degrade cartilage. Together they may slow cartilage breakdown, though research is mixed.

DOSE NOTES

1500 mg glucosamine sulfate + 1200 mg chondroitin sulfate daily. Glucosamine sulfate (not HCl) has stronger evidence.

TIMING NOTES

Split into 2–3 doses with meals to reduce GI upset.

GOOD FOR

- Knee osteoarthritis
- Joint aches from age or cumulative training
- Those who've tried NSAIDs and want an alternative

AVOID IF

- Shellfish allergy (most glucosamine comes from shellfish — look for vegetarian versions)
- You're on warfarin (chondroitin can increase INR)
- Uncontrolled diabetes (monitor blood sugar)

SIDE EFFECTS

- Mild GI upset
- Rare: allergic reaction
- May slightly raise blood glucose in some

COMMON FORMS

Glucosamine sulfate + chondroitin sulfate
Combination studied in major trials

Glucosamine HCl
Weaker evidence than sulfate form

Vegetarian glucosamine (from Aspergillus)
For shellfish-allergic users

STACKS WITH

col · omega3 · vitd

MAGNESIUM GLYCINATE

Calms the nervous system and supports deeper sleep.

DOSE	TIMING	ONSET
300–400 mg	Evening	Sleep benefits often felt within 1–3 nights; muscle cramp relief within a week

BENEFITS

- Promotes parasympathetic (rest-and-digest) nervous system activity
- Supports deeper, more restorative sleep
- Relaxes muscles — helps with cramps and tension
- Cofactor in 300+ enzyme reactions including ATP production

HOW IT WORKS

Magnesium is a natural NMDA receptor antagonist and GABA agonist — it calms excitatory neurotransmission. Glycinate (bound to the amino acid glycine) is gentle on the gut and crosses the blood-brain barrier well.

DOSE NOTES

300–400 mg elemental magnesium in the evening. Note: a '500 mg magnesium glycinate' capsule contains only ~100 mg elemental magnesium — check the label.

TIMING NOTES

30–60 minutes before bed. Split the dose if you get loose stools.

GOOD FOR

- Poor sleep or frequent waking
- Muscle cramps or restless legs
- High-stress periods
- Heavy training (magnesium is lost in sweat)

AVOID IF

- Kidney disease (talk to your doctor)
- You take bisphosphonates or certain antibiotics (separate by 2+ hours)
- You have low blood pressure and bradycardia

SIDE EFFECTS

- Loose stools (more common with oxide or citrate)
- Mild drowsiness
- Rare: low blood pressure at high doses

COMMON FORMS

Glycinate (bisglycinate)
Best for sleep and anxiety; gentle on gut

Citrate
Good absorption but laxative effect

Malate
Energy and muscle recovery

Oxide
Poorly absorbed — skip unless you need a laxative

Threonate
Crosses blood-brain barrier well; pricier

STACKS WITH

MELATONIN

Small dose helps nudge your sleep cycle without grogginess.

DOSE	TIMING	ONSET
0.3–1 mg	Evening	30–60 minutes after taking; phase shifts over several days

BENEFITS

- Signals your brain that it's time to sleep
- Shortens time-to-fall-asleep for delayed sleep phase
- Helpful for jet lag and shift work
- Small doses (0.3–1 mg) work as well or better than large ones

HOW IT WORKS

Melatonin is a hormone made by the pineal gland in response to darkness. Supplemental melatonin binds MT1/MT2 receptors and shifts the circadian phase — it's a timing cue, not a sedative.

DOSE NOTES

Start at 0.3–1 mg. Most store-bought brands sell 3–10 mg, which is 10x+ what's needed and often causes next-day grogginess. Less is more.

TIMING NOTES

30–60 minutes before target bedtime. For jet lag eastward, take at destination bedtime.

GOOD FOR

- Jet lag
- Shift work
- Delayed sleep phase (trouble falling asleep at a normal hour)
- Occasional difficulty falling asleep

AVOID IF

- You're pregnant or breastfeeding
- You have autoimmune disease (data is mixed)
- You're on blood thinners or immunosuppressants
- You wake up with grogginess — you're overdosing

SIDE EFFECTS

- Vivid dreams
- Morning grogginess (usually means the dose is too high)
- Headache
- Mild mood changes in some

COMMON FORMS

Immediate-release 0.3–1 mg
Ideal starting dose

Sublingual
Faster onset; useful for middle-of-night waking

Extended-release
For staying asleep, not falling asleep

STACKS WITH

PROBIOTIC (MULTI-STRAIN)

Supports a diverse gut microbiome.

DOSE	TIMING	ONSET
10–30 B CFU	Morning	2–8 weeks for most effects; strain-dependent

BENEFITS

- May reduce antibiotic-associated diarrhea
- Supports gut barrier and may reduce bloating in IBS
- Some strains help with traveler's diarrhea
- Specific strains studied for immune and mood effects

HOW IT WORKS

Probiotics are live bacteria that transiently colonize the gut, compete with pathogens, produce short-chain fatty acids, and modulate immune signaling. Effects are strain-specific — not all probiotics do the same thing.

DOSE NOTES

10–30 billion CFU is a reasonable range. More isn't necessarily better — strain selection matters more than dose.

TIMING NOTES

Morning with or without food depending on product. Enteric-coated or shelf-stable formulas are easier to time.

GOOD FOR

- Post-antibiotic recovery
- IBS, mild bloating, constipation
- Travelers (Saccharomyces boulardii)
- Poor diet diversity

AVOID IF

- Severe immunocompromise (discuss with doctor)
- Central venous catheter or critical illness

SIDE EFFECTS

- Transient gas or bloating in first week
- Rare: infection in immunocompromised patients

COMMON FORMS

Multi-strain capsule
Most general-purpose; refrigerated often needed

Saccharomyces boulardii
Yeast — good for antibiotic-associated diarrhea

Shelf-stable spore-based (Bacillus)
Survives stomach acid well; no refrigeration

Fermented foods
Yogurt, kefir, kimchi, sauerkraut — whole-food alternative

STACKS WITH

fiber | multi

FAQ

PROTEIN POWDER

Helps you hit daily protein targets for recovery and muscle growth.

DOSE	TIMING	ONSET
25–30 g	Afternoon	Daily protein adequacy matters more than any single shake; recovery benefits accumulate over weeks

BENEFITS

- Convenient way to hit daily protein targets (0.7–1 g/lb bodyweight for active people)
- High leucine content triggers muscle protein synthesis
- Faster digestion than whole-food protein post-workout
- Supports satiety and body composition

HOW IT WORKS

Dietary protein provides amino acids — the building blocks for muscle, enzymes, and hormones. Leucine in particular signals mTOR to initiate muscle protein synthesis. Whey is ~25% branched-chain amino acids and digests quickly.

DOSE NOTES

20–40 g per serving, 1–2x per day. It's a protein supplement, not a meal replacement — prioritize whole-food protein.

TIMING NOTES

Any time you need protein — post-workout, between meals, or as breakfast. Total daily protein matters most.

GOOD FOR

- Strength training / muscle gain
- Weight loss (protein preserves muscle, boosts satiety)
- Busy schedules with low-protein meals
- Older adults fighting sarcopenia

AVOID IF

- Whey: milk allergy or severe lactose intolerance (use isolate or plant)
- You already hit protein targets via food

SIDE EFFECTS

- Bloating or gas (lactose — switch to isolate or plant)
- Constipation if fiber intake stays low
- High doses stress kidneys only if kidneys are already compromised

COMMON FORMS

Whey concentrate
80% protein; cheapest; contains lactose

Whey isolate
90%+ protein; minimal lactose

Casein
Slow-digesting; good before bed

Plant (pea + rice blend)
Vegan option; blend for complete amino profile

PSYLLIUM HUSK

Prebiotic fiber feeds good gut bacteria and aids digestion.

DOSE	TIMING	ONSET
5 g	Evening	Days for bowel regularity; 4–8 weeks for cholesterol benefits

BENEFITS

- Improves both constipation and diarrhea (normalizes stool)
- Lowers LDL cholesterol (FDA health claim)
- Blunts post-meal blood sugar spikes
- Feeds beneficial gut bacteria — short-chain fatty acids

HOW IT WORKS

Psyllium is a soluble fiber that forms a gel in water. It slows gastric emptying, binds bile acids (forcing the liver to pull cholesterol from blood), and is fermented by colonic bacteria to produce butyrate.

DOSE NOTES

Start with 5 g/day and work up to 10–15 g split across meals. Drink plenty of water — essential to avoid obstruction.

TIMING NOTES

Before meals lowers post-meal glucose most. Evening works for bowel regularity.

GOOD FOR

- Low-fiber diets
- Constipation or loose stools
- Elevated LDL cholesterol
- Pre/post meal blood sugar management
- Keto dieters (often low fiber)

AVOID IF

- Bowel obstruction or severe swallowing difficulty
- You take certain medications — space by 2 hours (can reduce absorption)

SIDE EFFECTS

- Gas and bloating in first 1–2 weeks — ramp up slowly
- Choking risk if taken with too little water
- Reduced absorption of some medications taken simultaneously

COMMON FORMS

Powder
Cheapest; mix in water quickly and drink

Capsules
Convenient but need 5–6 to equal one powder dose

Whole husk vs powder
Powder is finer, mixes smoother

STACKS WITH

VITAMIN C

Antioxidant support and immune resilience.

DOSE	TIMING	ONSET
500 mg	Morning	Saturates tissues within days; benefits sustained with daily intake

BENEFITS

- Powerful water-soluble antioxidant
- Essential cofactor for collagen synthesis
- Enhances iron absorption from plant foods
- May modestly reduce cold duration

HOW IT WORKS

Ascorbic acid donates electrons to neutralize free radicals, regenerates vitamin E, and is required for enzymes that hydroxylate proline and lysine during collagen formation.

DOSE NOTES

200–1000 mg/day is the effective range. Above ~400 mg, absorption drops off steeply and the excess is excreted. Mega-dosing wastes money.

TIMING NOTES

Anytime. Pair with iron-rich plant meals to boost iron absorption.

GOOD FOR

- Low fruit/vegetable intake
- Smokers (need ~35 mg extra/day)
- Plant-based diets (to boost non-heme iron absorption)
- Periods of high physical or emotional stress

AVOID IF

- You have iron overload (hemochromatosis) — enhances iron absorption
- History of kidney stones at very high doses

SIDE EFFECTS

- GI upset and diarrhea above 2 g/day
- Slightly increased oxalate — relevant for stone formers

COMMON FORMS

Ascorbic acid
Cheapest, effective

Buffered (calcium/magnesium ascorbate)
Easier on sensitive stomachs

Liposomal
Higher absorption at large doses; pricier

STACKS WITH

zinc · iron · multi

ZINC

Supports immune cells, wound healing, and testosterone.

DOSE	TIMING	ONSET
15 mg	Morning	Cold duration benefits within days; long-term immune support builds over weeks

BENEFITS

- Supports innate and adaptive immune function
- May shorten duration of common cold if started within 24 hours
- Critical for wound healing and skin health
- Cofactor in testosterone production

HOW IT WORKS

Zinc is a cofactor in 300+ enzymes and stabilizes cell membranes. Deficiency impairs T-cell function and slows tissue repair.

DOSE NOTES

15–30 mg/day long-term. Short-term (cold onset): up to 75 mg/day lozenges for 5–7 days. Upper limit is 40 mg/day long-term — more can induce copper deficiency.

TIMING NOTES

With food to avoid nausea. Separate from calcium, iron, and coffee (they reduce absorption).

GOOD FOR

- Frequent colds
- Vegetarians/vegans (plant zinc is less bioavailable)
- Men supporting testosterone
- Slow wound healing

AVOID IF

- You take high-dose zinc already from a multi
- You're on certain antibiotics (separate by 2 hours)
- You have Wilson's disease

SIDE EFFECTS

- Nausea on empty stomach
- Metallic taste from lozenges
- Long-term high doses cause copper deficiency — anemia, neurological issues

COMMON FORMS

Picolinate
Well-absorbed

Bisglycinate
Gentle on stomach

Citrate
Good absorption, inexpensive

Lozenges (acetate/gluconate)
For cold onset — contact with throat tissue matters

Oxide
Poorly absorbed — skip

STACKS WITH

B-COMPLEX

Essential when skipping animal products — energy and nerve function.

DOSE	TIMING	ONSET
1 capsule	Morning	Days to weeks for energy, mood, and cognitive effects in deficient individuals

BENEFITS

- Supports energy metabolism — every B vitamin is a cofactor in ATP production
- Important for nerve function and neurotransmitter synthesis
- Folate (B9) essential for DNA synthesis and methylation
- Methylated forms benefit people with MTHFR variants

HOW IT WORKS

B vitamins are water-soluble cofactors in dozens of metabolic enzymes. They're not stored (except B12), so steady intake matters. Deficiencies in B1, B6, B9, B12 are the most common clinically.

DOSE NOTES

One capsule of a balanced B-complex daily. Avoid mega-B products (500+ mg of each) — no added benefit and some cause side effects.

TIMING NOTES

Morning with food. Some people find it energizing — avoid late-day dosing.

GOOD FOR

- Vegans (B12 gap)
- High-stress periods (Bs deplete faster)
- Alcohol use (impairs B1/thiamine)
- MTHFR variants (methylated forms)

AVOID IF

- You already take high-dose individual B vitamins
- You have a rare Leber's hereditary optic neuropathy (avoid cyanocobalamin)

SIDE EFFECTS

- Bright yellow urine (harmless — B2)
- Flushing from high-dose niacin (B3)
- Neuropathy from chronic high-dose B6 (>200 mg/day long-term)

COMMON FORMS

Standard B-complex
Balanced, affordable

Methylated B-complex
Methylfolate + methylcobalamin — preferred for MTHFR

Activated B-complex
All Bs in coenzyme forms; pricier

STACKS WITH

b12 · iron · multi

ELECTROLYTES (NA/K/MG)

Keto and heavy training deplete sodium and potassium quickly.

DOSE	TIMING	ONSET
1 scoop	Afternoon	Immediate — often felt within 15–30 minutes if depleted

BENEFITS

- Replaces sodium and potassium lost in sweat
- Helps prevent exercise-induced cramps
- Supports hydration — water alone isn't always enough
- Critical on keto and low-carb diets (kidneys excrete more sodium)

HOW IT WORKS

Electrolytes (Na+, K+, Mg++, Cl-) maintain cell membrane potential, fluid balance, and nerve/muscle firing. Sweating, low-carb diets, and hot weather all accelerate losses.

DOSE NOTES

One scoop/packet per workout or per liter of water on low-carb/hot days. Look for ~1000 mg sodium, 200+ mg potassium, 60+ mg magnesium per serving.

TIMING NOTES

Pre-workout, during long training, or first thing on low-carb days. Not needed if eating a balanced diet and not sweating heavily.

GOOD FOR

- Heavy training (>1 hour or high sweat)
- Keto or low-carb
- Hot climates or saunas
- Fasting periods

AVOID IF

- You have high blood pressure sensitive to sodium
- Kidney disease (talk to your doctor about potassium)
- You already eat a high-sodium diet and don't sweat much

SIDE EFFECTS

- Too much sodium — bloating, elevated BP in sensitive people
- High potassium rare from supplements but dangerous in kidney disease
- GI upset from magnesium

COMMON FORMS

Powder / stick packs
Flexible dosing; many flavors

Tablets
Convenient for travel

DIY (salt + lite salt + water)
Cheapest; no flavoring

STACKS WITH

creatine · whey · zinc

IRON BISGLYCINATE

Plant-based diets often run low on bioavailable iron.

DOSE	TIMING	ONSET
18 mg	Afternoon	Ferritin rises over 2–3 months; energy improvements often noticed within 2–4 weeks

BENEFITS

- Corrects iron-deficiency anemia
- Restores energy in deficient individuals
- Supports oxygen transport and cognitive function
- Bisglycinate form is gentler on the gut than ferrous sulfate

HOW IT WORKS

Iron is central to hemoglobin, which carries oxygen in red blood cells. Deficiency impairs oxygen delivery — fatigue, poor exercise tolerance, brain fog.

DOSE NOTES

18 mg bisglycinate daily for maintenance. Treating deficiency: 30–60 mg elemental iron under medical guidance, with ferritin monitoring. Don't supplement without confirmed deficiency.

TIMING NOTES

Between meals with vitamin C for best absorption. Avoid coffee, tea, calcium, and dairy within 2 hours — all block absorption.

GOOD FOR

- Menstruating women (especially heavy periods)
- Vegans and vegetarians (plant iron is less bioavailable)
- Endurance athletes
- Pregnancy (under medical guidance)

AVOID IF

- You haven't confirmed deficiency with a ferritin test
- You have hemochromatosis
- You're a man with normal iron (men rarely need to supplement)

SIDE EFFECTS

- Constipation (less common with bisglycinate)
- Nausea
- Dark stools (harmless)
- Iron overload with unnecessary supplementation — serious

COMMON FORMS

Bisglycinate
Gentlest; well-absorbed

Ferrous sulfate
Cheap, effective, more GI side effects

Ferrous gluconate

Heme iron

VITAMIN B12 (METHYLCOBALAMIN)

B12 is virtually absent in plant foods — non-negotiable for vegans.

DOSE	TIMING	ONSET
500 mcg	Morning	Weeks to months to correct deficiency; energy and cognitive benefits can be noticed earlier in deficient individuals

BENEFITS

- Essential for red blood cell formation — prevents anemia
- Required for myelin sheath — nerve function and cognition
- Supports methylation (DNA synthesis, homocysteine metabolism)
- Deficiency causes irreversible nerve damage if prolonged

HOW IT WORKS

B12 (cobalamin) is a cofactor for methionine synthase and methylmalonyl-CoA mutase. Only produced by bacteria — found almost exclusively in animal foods or fortified products.

DOSE NOTES

500–1000 mcg/day methylcobalamin or cyanocobalamin for vegans and vegetarians. Deficient individuals may need higher doses or injections under medical guidance.

TIMING NOTES

Morning. Sublingual absorbs without needing intrinsic factor.

GOOD FOR

- Vegans — non-negotiable
- Vegetarians — recommended
- Adults over 50 (reduced absorption from food)
- People on metformin or PPIs (both reduce B12 absorption)

AVOID IF

- Leber's disease (rare eye condition)
- You should confirm deficiency with bloodwork before injections

SIDE EFFECTS

- Essentially none — excess is excreted
- Rare: acne flare, anxiety at very high doses

COMMON FORMS

Methylcobalamin
Active form; preferred by some

Cyanocobalamin
Cheapest, stable, well-researched

Hydroxocobalamin
Used in injections

Sublingual
Absorbs without intrinsic factor

VITAMIN D3 + K2

Critical when sun exposure is limited; supports mood and bones.

DOSE	TIMING	ONSET
2000–4000 IU	Morning	4–8 weeks to raise serum 25(OH)D; get a blood test before and after

BENEFITS

- Supports bone density by regulating calcium absorption
- Modulates immune function; deficiency linked to frequent infections
- May improve mood, especially in winter months
- K2 directs calcium into bones and away from arteries

HOW IT WORKS

Vitamin D3 (cholecalciferol) is converted to calcidiol in the liver, then calcitriol in the kidneys — a hormone that regulates hundreds of genes. K2 (MK-7) activates osteocalcin, which binds calcium into bone matrix.

DOSE NOTES

2000–4000 IU/day is a common maintenance range for adults with limited sun. Deficient individuals may need 5000+ IU under medical guidance. Target serum 25(OH)D of 30–50 ng/mL.

TIMING NOTES

Take with the largest fatty meal of the day — it's fat-soluble.

GOOD FOR

- Limited sun exposure (indoor work, northern latitudes, dark skin)
- Over 50 (skin synthesizes less efficiently)
- Bone density concerns
- Frequent colds

AVOID IF

- You have hypercalcemia or sarcoidosis
- You're on high-dose warfarin (K2 interacts)
- You take thiazide diuretics (monitor calcium)

SIDE EFFECTS

- Rare at sensible doses
- Hypercalcemia with extreme doses (10,000+ IU/day sustained)
- Nausea or constipation from excess

COMMON FORMS

D3 (cholecalciferol)
Preferred — better raises blood levels than D2

D3 + K2 (MK-7) combo
Synergistic for bone and arterial health

Liquid drops
Flexible dosing; good for kids

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